Care Access Navigator

Incident Response 1 — Task 2: Incident Detection and Analysis

Parrot OS and macOS log analysis with Wazuh, a correlated suspicious-login investigation, and severity classification of three incidents

Document ID	CAN-IR1-T2-LOG-002 (replaces version 001: same lab, people and incident numbers as Tasks 1 and 3–6)
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SIEM	Wazuh manager 4.9.2 + agent on parrot-vm (Task 1); 15 custom rules, 3 CDB lists from the JML register
Hosts	**Parrot OS**: parrot-vm (agent 001). **macOS**: DAEDMAC1009, the analyst's Mac (install.log + SSH client trail)
People	Sathya Hasi (joiner), Tara Jones (mover), Jeevan Smith (leaver), it_admin; patient records are synthetic
Run	scenario 19:29:50 UTC; Wazuh SEV-1 alert 19:31:10 UTC; analysis done 19:37:18 UTC, 2026-11-03

Summary

The JML people acted on parrot-vm from the Mac, and Wazuh analysed both hosts' logs. It raised 1,683 alerts: 268 from the Parrot agent and 1,225 from macOS. 23 were at level 10 or higher. Every high-level Parrot alert was raised within 1.9 s of the event it reports. Three incidents came out of it, and each one maps to a JML failure. Tara Jones (mover) read a patient record through a group she should have lost: rule 100221 fired at 19:31:10 and started containment (Task 3). Sathya Hasi (joiner) was given sudo with no change record and used root. Jeevan Smith (leaver) tried to log in twice. His attempts were investigated end to end and matched across the Mac's SSH client, a packet capture, the Parrot sshd log and Wazuh. The severity matrix rates them **SEV-1** (IR1-2026-002), **SEV-2** (IR1-2026-003) and **SEV-3** (IR1-2026-001). The macOS analysis found patch lag, a remote-access tool whose approval is not yet confirmed, and a recurring false-positive pattern, with a tuning proposal.

1. How this report meets the Task 2 requirements

Requirement	Where	Evidence
Parrot OS and macOS logs analysed with Wazuh	§4, §5	wazuh/alerts.json, analysis/alerts_by_rule.csv
Methodology documented	§3	lab/wazuh/*.xml, lists/
1 suspicious login: correlation host ↔ Parrot	§6.1	analysis/correlation_ir1_2026_001.csv, pcap/
Event timeline	§6.2	analysis/timeline_ir1_2026_001.csv
Alert validation process	§6.3	transcripts/06, 08

Requirement	Where	Evidence
3 incidents classified with the severity matrix	§7	§7.2
Security implications	§4.2, §5.2, §8	

2. Data collected

Source	Host	Collection	Received
`/var/log/auth.log`, `ehr-audit.log`, `kern.log`, `audit.log` + FIM	Parrot	Wazuh agent 001 (live)	268 alerts
`/var/log/install.log`	macOS	file shipped to a manager input (Task 1 proved the path)	51,823 / 51,823 lines
`ssh_client.log` (OpenSSH client trail)	macOS	file shipped	578 / 578 lines
`ssh_sessions.pcapng`	Parrot	tshark, saved filter 'IR1 SSH sessions'	client banners

transcripts/02_scenario.txt - what the people did (from the Mac)

```
[2026-11-03T19:29:50Z] S1 Sathya Hasi (test.clinician, joiner) views her patient PT-1001
SYNTHETIC RECORD PT-1001

[2026-11-03T19:30:01Z] S2 it_admin adds test.clinician to the sudo group (no change ticket)
uid=1002(test.clinician) gid=1005(test.clinician) groups=1005(test.clinician),27(sudo),1001(clinical)

[2026-11-03T19:30:13Z] S3 Sathya Hasi runs a command as root
uid=0(root) gid=0(root) groups=0(root)

[2026-11-03T19:30:30Z] S4a login attempt as Jeevan Smith (test.billing, leaver)

[2026-11-03T19:30:51Z] S4b second login attempt as test.billing
[2026-11-03T19:31:06Z] >>> s5_start

[2026-11-03T19:31:06Z] S5a Tara Jones (test.nurse, now Billing) logs in and checks her groups
uid=1003(test.nurse) gid=1006(test.nurse) groups=1006(test.nurse),1001(clinical),1002(billing)
parrot-vm

[2026-11-03T19:31:07Z] S5b opens clinical record PT-1003 and stages a copy in a hidden folder
total 12
drwxr-xr-x 2 test.nurse test.nurse 4096 Nov  3 19:31 .
drwxr-xr-x 3 test.nurse test.nurse 4096 Nov  3 19:31 ..
-rw-r--r-- 1 test.nurse test.nurse  127 Nov  3 19:31 pt1003.txt

[2026-11-03T19:31:08Z] S5c drops syncd.py (uploader) next to it
9f3dbfe2b4d8cf2200bd95b6d08a2cab4336d74b2bb15d3adc02369b5cfe3477  /home/test.nurse/.cache/.sync/syncd.py

[2026-11-03T19:31:09Z] S5d starts it detached from the session
1534 python3 /home/test.nurse/.cache/.sync/syncd.py

[2026-11-03T19:31:11Z] S5e adds a cron job that restarts it (persistence)
*/5 * * * * pgrep -f syncd.py >/dev/null || setsid python3 $HOME/.cache/.sync/syncd.py >/dev/null 2>&1 &
```

3. Analysis method

#	Step	What was done
1	Collect	Parrot logs live through the agent; macOS logs shipped; sent = received checked for every file
2	Decode	Wazuh pre-decoders, bundled decoders and the custom ones (ehr-audit, usermod-group, macOS)
3	Detect	Bundled rules (sshd, sudo, PAM, FIM, audit) + 15 custom rules that look identities up in the JML CDB lists
4	Triage	Sort by level; the SEV-1 alert (100221) goes straight to containment, the rest queue by severity
5	Validate	Five checks per high alert: raw log re-read, wazuh-logtest replay, cross-host corroboration, business context (JML ticket), false-positive test
6	Correlate	Mac ↔ Parrot by time, SSH client banner, TCP port (pcap ↔ sshd) and outcome
7	Timeline	All sources in UTC, plus the JML register events
8	Classify	Impact × Urgency matrix → SEV-1–4 (playbook CAN-IR-PB-02 §0)

4. Parrot OS log analysis

4.1 Alerts (Parrot, level 5 and above)

Rule	Lvl	Description	Count	First (UTC)
100221	12	ePHI viewed by mover test.nurse (now Billing) through stale clinical group - rec	1	19:31:08
100231	12	Root obtained via sudo by test.clinician, whose role has no root entitlement (RB	1	19:30:15
100211	11	Repeated login attempts by JML leaver account test.billing (2+ in 10 min)	1	19:30:51
100210	10	JML leaver account test.billing attempted SSH login from 10.78.10.1 - account wa	1	19:30:31
100230	10	Account test.clinician added to privileged group sudo	2	19:30:03
80710	10	Auditd: Device enables promiscuous mode.	17	19:22:02
5104	8	Interface entered in promiscuous(sniffing) mode.	9	19:22:02
550	7	Integrity checksum changed.	4	19:21:59
553	7	File deleted.	3	19:31:11
554	5	File added to the system.	8	19:21:59

Triage queue when the SEV-1 fired (transcripts/03)

```
# alerts raised so far by the IR1 rules (triage queue, highest level first)
19:30:17.751 L12 100231 Root obtained via sudo by test.clinician, whose role has no root entitlement
    (RBAC/SoD vio
19:31:10.008 L12 100221 ePHI viewed by mover test.nurse (now Billing) through stale clinical group -
    record PT-100
19:30:51.868 L11 100211 Repeated login attempts by JML leaver account test.billing (2+ in 10 min)
19:30:03.638 L10 100230 Account test.clinician added to privileged group sudo
19:30:03.641 L10 100230 Account test.clinician added to privileged group sudo
19:30:31.816 L10 100210 JML leaver account test.billing attempted SSH login from 10.78.10.1 - account
    was deprovis
19:24:32.614 L8  100244 macOS host: remote-access tool installed - Splashtop Streamer (validate against
    approved s
19:23:22.465 L5  100241 macOS host: OS update offered but not installed - macOS Tahoe 26.7.1 25G241
19:23:22.465 L5  100241 macOS host: OS update offered but not installed - macOS 27.0.1 26A434
    (Customer)(SU)
19:23:22.467 L5  100241 macOS host: OS update offered but not installed - macOS 27.0.1 26A434
    (Customer)(SU)
19:23:22.467 L5  100241 macOS host: OS update offered but not installed - macOS Tahoe 26.7.1 25G241
19:23:22.467 L5  100241 macOS host: OS update offered but not installed - macOS 27.0.1 26A434
    (Customer)(SU)
19:23:22.467 L5  100241 macOS host: OS update offered but not installed - macOS Tahoe 26.7.1 25G241
19:23:22.467 L5  100241 macOS host: OS update offered but not installed - macOS 27.0.1 26A434
    (Customer)(SU)
19:23:22.467 L5  100241 macOS host: OS update offered but not installed - macOS Tahoe 26.7.1 25G241
```

4.2 Findings and implications (Parrot)

ID	Finding	Implication
P-1	Rule 100221 (L12) at 19:31:10: test.nurse, now Billing, viewed PT-1003 through group clinical. The FIM alerts that followed show files appearing in a hidden folder in her home and a new crontab.	ePHI exposed through privilege creep; the files and cron entry point to staging and persistence. SEV-1 → containment started at the alert (Task 3).
P-2	Rules 100230 + 100231 at 19:30:03–19:30:15: it_admin added test.clinician to sudo, then she ran a command as root.	Root for a role with no root entitlement and no change record: a least-privilege and SoD breach on an ePHI host.
P-3	Rules 100210 + 100211 from 19:30:31: leaver test.billing tried SSH twice; sshd refused (DenyGroups).	Blocked, but the leaver's SSH key is still on the host (investigated in §6).

5. macOS log analysis

Source: DAEDMAC1009's `/var/log/install.log` (51,823 lines from 2026-10-13) and its SSH client trail. The unified log needs admin rights on this managed Mac, so it could not be read.

Rule	Lvl	Description	Count
100244	8	macOS host: remote-access tool installed - Splashtop Streamer (validate against approved s	1
100251	7	macOS host: SSH client used leaver account test.billing against 127.0.0.1:2223	2
100241	5	macOS host: OS update offered but not installed - macOS Tahoe 26.7.1 25G241	342
100241	5	macOS host: OS update offered but not installed - macOS 27.0.1 26A434 (Customer)(SU)	342
100241	5	macOS host: OS update offered but not installed - macOS Tahoe 26.7.2 25G332	153
100241	5	macOS host: OS update offered but not installed - macOS 27.2 26B123 (Customer)(SU)	153
100242	5	macOS host: process 75876 tried to change Software Update settings without entitlement	77
100240	3	macOS host power state: system has powered on	70
100240	3	macOS host power state: system will sleep	69
100250	3	macOS host: SSH client connecting to 127.0.0.1:2223 as test.clinician	2
100250	3	macOS host: SSH client connecting to 127.0.0.1:2223 as test.nurse	5
100252	3	macOS host: SSH session refused or closed by the remote host	2
100253	3	macOS host: SSH client authenticated to the remote host	7

5.1 Host-side validation

transcripts/08_macos_validation.txt

```
# patch level: installed version vs the updates install.log says were offered
$ sw_vers
ProductName:        macOS
ProductVersion:     26.3.1
BuildVersion:       25D2128

# remote-access software seen in install.log: is it present, who signed it, does Gatekeeper accept it?
$ pkgutil --pkgs | grep -i splashtop
com.splashtop.soundDriver
com.splashtop.Splashtop-Streamer
$ codesign -dv --verbose=2 "/Applications/Splashtop Streamer.app"
Identifier=com.splashtop.Splashtop-Streamer
Authority=Developer ID Application: Splashtop Inc. (CPQQ3AW49Y)
Authority=Developer ID Certification Authority
Authority=Apple Root CA
Timestamp=Sep 24, 2026 at 1:44:37AM
TeamIdentifier=CPQQ3AW49Y
$ spctl -a -vv "/Applications/Splashtop Streamer.app"
/Applications/Splashtop Streamer.app: accepted
source=Notarized Developer ID
origin=Developer ID Application: Splashtop Inc. (CPQQ3AW49Y)

# MDM enrolment (managed device => IT-deployed software is plausible)
$ profiles status -type enrollment
Enrolled via DEP: Yes
MDM enrollment: Yes (User Approved)
MDM server: https://biz-30445.business.mosyle.io/mdm/server/?op=commands&lang=en&c=234230350111783f7a11&
    lifetime=1773414327

# processes behind the Software Update entitlement alerts (rule 100242), if still running
PIDs: 75876
75876 (not running now)
```

5.2 Findings and implications (macOS)

ID	Finding	Validation	Implication
M-1	Running macOS 26.3.1; updates offered and not installed: macOS Tahoe 26.7.1 25G241 (342×); macOS 27.0.1 26A434 (Customer)(SU) (342×); macOS Tahoe 26.7.2 25G332 (153×); macOS 27.2 26B123 (Customer)(SU) (153×)	True positive: sw_vers = 26.3.1	Missing security fixes on a clinical workstation; patch and plan the major upgrade
M-2	Remote-access tool installed: Splashtop Streamer 2026-10-16	Signature and Gatekeeper result in §5.1	Persistent inbound remote-control path (T1219): confirm with IT that it is approved, MFA-protected and logged
M-3	77 Software Update 'not entitled' events (PIDs 75876)	Process check in §5.1	Benign app behaviour seen before (Safari): tune by validated process, not by ignoring the rule
M-4	69 sleep / 70 wake events	Context	The host was in use when the scenario sessions left it
M-5	Unified log not readable; no agent on the Mac	transcript 04 (Task 1)	Mac authentication is invisible to the SIEM: deploy the agent through MDM

6. Investigation: suspicious login (IR1-2026-001)

Two SSH logins as Jeevan Smith (test.billing), terminated on TERM-2026-0301, came from the analyst workstation. The Mac sees its own connection to 127.0.0.1:2223; parrot-vm sees the Docker gateway. The attempts were therefore matched on time, the client's SSH banner, the TCP source port (pcap ↔ sshd) and the outcome on both sides.

6.1 Correlation across hosts

Attempt	Mac (ssh_client.log)	Network (pcap)	Parrot sshd	Wazuh	Δ Mac→sshd
1	19:30:30 pid 48999 Permission denied (publickey)	19:30:30.964 10.78.10.1:55708	19:30:31.159 sshd[1467] User test.billing from 10.78.10.1 not allowed because a grou	100210 (L10) 19:30:31.816	+1.11 s
2	19:30:51 pid 49063 Permission denied (publickey)	19:30:51.244 10.78.10.1:61926	19:30:51.420 sshd[1469] User test.billing from 10.78.10.1 not allowed because a grou	100211 (L11) 19:30:51.868	+0.37 s

6.2 Timeline

UTC	Host	Source	Event
10-19 18:13	Entra ID	JML-LOG TERM-2026-0301	Jeevan Smith (test.billing) disabled in Entra ID (leaver)
10-19 18:15	Entra ID	JML-LOG TERM-2026-0301	Sessions revoked; app role removed 18:17
10-19 ~18:28	Keycloak	JML-LOG TERM-2026-0301	Keycloak copy disabled
11-03 19:21:04.020	Parrot	auth.log usermod	Lab replay of leaver: change user 'test.billing' expiration from 'never' to '1970-01-02'
11-03 19:21:04.021	Parrot	auth.log usermod	Lab replay of leaver: add 'test.billing' to group 'disabled-users'
11-03 19:30:30Z	macOS	ssh_client.log ssh[48999]	Attempt 1: Mac user 'Adult' starts ssh as test.billing to 127.0.0.1:2223 (SSH-2.0-OpenSSH_10.2)
11-03 19:30:30.964	Network	ssh_sessions.pcapng	Attempt 1: client banner SSH-2.0-OpenSSH_10.2 from 10.78.10.1:55708
11-03 19:30:31.109	Parrot	auth.log sshd[1467]	Attempt 1: Connection from 10.78.10.1:55708 (Docker gateway for 127.0.0.1:2223)
11-03 19:30:31.159	Parrot	auth.log sshd[1467]	Attempt 1: User test.billing from 10.78.10.1 not allowed because a group is listed in DenyGroup
11-03 19:30:31.816	Wazuh	rule 100210 (L10)	Alert raised from the Wazuh agent on parrot-vm
11-03 19:30:31Z	macOS	ssh_client.log ssh[48999]	Attempt 1: client receives Permission denied (publickey)

UTC	Host	Source	Event
11-03 19:30:51Z	macOS	ssh_client.log ssh[49063]	Attempt 2: Mac user 'Adult' starts ssh as test.billing to 127.0.0.1:2223 (SSH-2.0-OpenSSH_10.2)
11-03 19:30:51.244	Network	ssh_sessions.pcapng	Attempt 2: client banner SSH-2.0-OpenSSH_10.2 from 10.78.10.1:61926
11-03 19:30:51.374	Parrot	auth.log sshd[1469]	Attempt 2: Connection from 10.78.10.1:61926 (Docker gateway for 127.0.0.1:2223)
11-03 19:30:51.420	Parrot	auth.log sshd[1469]	Attempt 2: User test.billing from 10.78.10.1 not allowed because a group is listed in DenyGroup
11-03 19:30:51.868	Wazuh	rule 100211 (L11)	Alert raised from the Wazuh agent on parrot-vm
11-03 19:30:51Z	macOS	ssh_client.log ssh[49063]	Attempt 2: client receives Permission denied (publickey)
11-03 19:36:44.335	Wazuh	rule 100251 (L7)	Attempt 1: macOS-side alert when the host log was shipped
11-03 19:36:44.336	Wazuh	rule 100251 (L7)	Attempt 2: macOS-side alert when the host log was shipped

6.3 Alert validation

wazuh-logtest -v replay of the raw sshd denial (transcripts/06)

```
        *Rule 1 matched
        *Rule 5700 matched
        *Rule 5718 matched
        *Rule 100210 matched
    id: '100210'
    level: '10'
    description: 'JML leaver account test.billing attempted SSH login from 10.78.10.1 - account was
    deprovisioned'
**Alert to be generated.
```

Check	Result
Raw log re-read	auth.log line 'User test.billing ... not allowed because a group is listed in DenyGroups' present for both attempts
Replay	wazuh-logtest reaches 5718 → 100210 (above): the alert follows from the log alone
Cross-host	Mac shows 2 refused sessions at the same seconds (§6.1)
Business context	JML-LOG TERM-2026-0301: Jeevan Smith disabled in Entra ID and Keycloak; no reinstatement ticket
False positive?	No: the account is on jml-leavers and was used. But the SSH key is still authorized on the host, so only one sshd setting stood in the way

7. Classification with the severity matrix

Impact × Urgency, each 1–3. The scores follow the prioritization factors in NIST SP 800-61r2 §3.2.6, and the product maps to the playbook's SEV-1–4 levels (CAN-IR-PB-02 §0). If the course's own matrix differs, only this mapping changes.

I \ U	U = 1	U = 2	U = 3
I = 3	3 → SEV-3	6 → SEV-2	9 → **SEV-1**
I = 2	2 → SEV-4	4 → SEV-3	6 → SEV-2
I = 1	1 → SEV-4	2 → SEV-4	3 → SEV-3

7.2 The three incidents

Incident	Who / JML	I	U	Severity	Rationale
IR1-2026-002 ePHI read via stale role	Tara Jones, mover CHG-2026-0201; 100221 at 19:31:10	3	3	**SEV-1** (9)	Patient record read by a Billing employee; FIM shows staging and persistence. Access still open: disclosure likely in progress
IR1-2026-003 Unapproved root	Sathya Hasi, joiner REQ-2026-0101; it_admin; 100230/100231	3	2	**SEV-2** (6)	Root on an ePHI host for a role without entitlement, no change record; used once (id), no data touched
IR1-2026-001 Leaver login attempts	Jeevan Smith, leaver TERM-2026-0301; 100210/100211	2	2	**SEV-3** (4)	Blocked, so no access; but it exposed a live gap (key still authorized) and was repeated

> **What happened next:** The SEV-1 did not wait for this analysis: containment started at the alert (Task 3), and this classification confirmed it afterwards. The SEV-2 and SEV-3 were queued and contained in the same session (Task 3, C7). Recovery and the root-cause fix for all three are in Task 6.

8. Security implications

- **One root cause, three incidents.** JML changes reached Entra ID and Keycloak but not the Linux host's own groups and keys.
- **Identity context is what made the alerts loud.** Bundled rules scored these events 3–5. The JML lists raised them to 10–12.
- **NAT hides the source.** Without the Mac's own log and the packet capture, the leaver attempts end at the Docker gateway.
- **Endpoint hygiene.** Patch lag and an unconfirmed remote-access tool on the analyst's Mac widen the attack surface.

Appendix A — Evidence index

evidence/EVIDENCE-INDEX-SHA256.txt

```
845ff23fa91c370b...  ./analysis/alerts_by_rule.csv
dd6dad2d7c52bf02...  ./analysis/correlation_ir1_2026_001.csv
f7028c66a3fe71ca...  ./analysis/detection_latency.csv
4948ea5b2f71b7e7...  ./analysis/key_events.json
9b140650714a3167...  ./analysis/macos_findings.json
66a6368a78e88960...  ./analysis/timeline_all.csv
166ba80efaddc144...  ./analysis/timeline_ir1_2026_001.csv
79ae0bad9b0b2bff...  ./macos/install.log
3b023d79493b49ba...  ./macos/ssh_client.log
18b4fcfc7f2156fd...  ./markers.csv
e7baec5bbcaf6e79...  ./parrot/auth.log
21f89e60217f451a...  ./parrot/ehr-audit.log
25268ee9765e53b6...  ./pcap/client_banners.csv
1cc4e2deda643f3e...  ./pcap/ssh_sessions.pcapng
89763d0cf59278d0...  ./transcripts/01_rules_and_capture.txt
fa301eae222f3cd1...  ./transcripts/02_scenario.txt
7843d030a562e1cb...  ./transcripts/03_detection.txt
90daab179b67a9ca...  ./transcripts/04_macos_logs.txt
2d98ad04470bdf99...  ./transcripts/05_export.txt
5c0b6d793c7088d4...  ./transcripts/06_logtest_trace.txt
a5d3b823994333ee...  ./transcripts/07_analysis.txt
ef58e79b2c518624...  ./transcripts/08_macos_validation.txt
8f44f43e075a5366...  ./wazuh/alerts.json
f113d373dfddff98...  ./wazuh/manager_ossec.conf
```